Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Fine Tuning Open Source LLM

Course Code: C922

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 October 2026	Initial release of the Fine Tuning Open Source LLM lesson plan: one 9:30 am – 5:30 pm day (7.5 instructional hours) covering all four Learning Units and 15 topics, with six labs run in class and 20 self-paced extension labs, mapped to the v1.0 183-slide deck.	Tertiary Infotech Academy

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Course Information

Course Title	Fine Tuning Open Source LLM
Course Code	C922
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 7.5 instructional hours
Daily Timing	9:30 am – 5:30 pm (7.5 instructional hours). The single 30-minute lunch break is excluded from instructional time; refreshments are taken during activity time.
Mode of Learning	Physical, instructor-led, hands-on fine-tuning labs on Google Colab (T4 GPU)
Class Size	Maximum 20 learners
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply transformation methods to ingest and prepare synthetic data using cloud-based model tools.
- LO2: Develop optimized data pipelines for feature engineering using various optimization strategies.
- LO3: Fine-tune pre-trained multi-modal models using advanced loss metrics and training strategies.
- LO4: Analyse ML solutions for bias, explainability, and alignment with performance benchmarks.

Learning Units and Topics

Learning Unit	Topics	Learning Outcome	Hands-on labs
LU1: Data Engineering	Topic 1: Create data repositories for fine tuning Topic 2: Identify and implement a dataset ingestion solution Topic 3: Identify and implement a data transformation solution	LO1	Labs 1, 5 in class · Labs 2–4, 6 self-paced
LU2: Exploratory Data Analysis	Topic 1: Sanitize and prepare data for modelling Topic 2: Perform feature engineering for fine tuning Topic 3: Analyze and visualize data for fine tuning insights	LO2	Lab 7 in class · Labs 8 –12 self-paced
LU3: Modelling	Topic 1: Frame business problems as fine-tuning problems	LO3	Labs 15, 17 in class · Labs 13, 14, 16, 18–20 self-paced

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	Topic 2: Select the appropriate model and adaptation method Topic 3: Train and fine-tune models with LoRA, QLoRA and Unsloth Topic 4: Perform hyperparameter optimization Topic 5: Evaluate fine-tuned models		
LU4: Machine Learning Implementation and Operations	Topic 1: Build and benchmark generative solutions against state of the art Topic 2: Recommend and implement bias mitigation and fairness interventions Topic 3: Apply interpretability and explainability methods Topic 4: Deploy and operationalize fine-tuned models	LO4	Lab 21 in class · Labs 22–26 self-paced

Instructional Methods and Resources

Instructional methods	Interactive presentation; demonstration and modelling; guided hands-on practice; drill and practice; discussion and Q&A; case-based problem solving.
Learner resources	Trainer slides (PDF), Learner Guide, Lesson Plan, and 26 lab folders each containing a step-by-step README, a runnable script and its own mock dataset, or the output of an earlier lab in the same Learning Unit. Labs 1, 5, 7, 15, 17 and 21 run in class; the other 20 are self-paced extension work.
Training resources	Laptop per learner with a modern browser; Wi-Fi; TV/projector; whiteboard; Google Colab account with a free T4 GPU runtime.
Software	Google Colab, Python 3.11+, PyTorch, Hugging Face transformers / datasets / peft / trl / accelerate, bitsandbytes, Unsloth, evaluate.

Course Schedule

Day 1 • 9:30am – 5:30pm

Time	Duration	Instructions for Facilitator	Instructional Methods	Resources
0930hrs–0945hrs	15 mins	Welcome, trainer and learner introductions,	Interactive presentation, Discussion	Trainer slides, TV, Whiteboard

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		course structure and ground rules		
0945hrs–1005hrs	20 mins	Course orientation: learning outcomes and the one-day course map. Core concepts: why fine tuning; pre-training vs adaptation; prompt engineering vs RAG vs fine tuning; the adaptation decision tree (from slide 12).	Interactive presentation, Demonstrations / Modelling	Trainer slides, TV, Whiteboard, Wi-Fi
1005hrs–1145hrs	100 mins	LU1 — Data Engineering · Topic 1: Create data repositories for fine tuning — versioned repositories and synthetic data generation. Topic 2: Identify and implement a dataset ingestion solution. Topic 3: Identify and implement a data transformation solution — chat templates and the EOS token. Hands-on (in class): Lab 1: Set Up the Fine-Tuning Environment on Colab (slide 45); Lab 5: Transform Data into Chat and Instruction Formats (slide 57). Self-paced extension: Labs 2, 3, 4 and 6.	Demonstrations / Modelling, Drill and Practice	Trainer slides, Learner Guide, lab folder, Colab GPU, Wi-Fi
1145hrs–1225hrs	40 mins	LU2 — Exploratory Data Analysis · Topic 1: Sanitize and prepare data for modelling — PII, duplicates, near-duplicates, prompt injection, quality scoring and the ETL quality gate. Hands-on (in class): Lab 7: Clean and Sanitize a Fine-Tuning Dataset (slide 78).	Demonstrations / Modelling, Drill and Practice	Trainer slides, Learner Guide, lab folder, Colab GPU, Wi-Fi
1225hrs–1255hrs	30 mins	Lunch break	N/A	N/A
1255hrs–1355hrs	60 mins	Topic 2: Perform feature engineering for fine tuning — system prompts, exemplar selection, length bucketing,	Interactive presentation, Demonstrations / Modelling	Trainer slides, TV, Whiteboard, Wi-Fi

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		augmentation. Topic 3: Analyze and visualize data for fine tuning insights; decoding parameters and the generative concepts transfer learning, distillation and self-supervised learning (trainer demonstration). Self-paced extension: Labs 8–12.		
1355hrs–1455hrs	60 mins	LU3 — Modelling · Topic 1: Frame business problems as fine-tuning problems, multi-modal embeddings and models. Topic 2: Select the appropriate model and adaptation method. Topic 3: Train and fine-tune models with LoRA, QLoRA and Unsloth — LoRA from first principles, PEFT LoRA and QLoRA 4-bit quantisation. Hands-on (in class): Lab 15: Fine-Tune with PEFT LoRA (slide 121).	Demonstrations / Modelling, Drill and Practice	Trainer slides, Learner Guide, lab folder, Colab GPU, Wi-Fi
1455hrs–1555hrs	60 mins	Topic 3 continued: accelerating with Unsloth. Topic 4: Perform hyperparameter optimization — monitoring training stability and convergence, and the hyperparameter priority order. Topic 5: Evaluate fine-tuned models — advanced loss functions, generative metrics and intermediate fine tuning. Hands-on (in class): Lab 17: Accelerate Fine Tuning with Unsloth (slide 127). Self-paced extension: Labs 13, 14, 16, 18, 19 and 20.	Demonstrations / Modelling, Drill and Practice	Trainer slides, Learner Guide, lab folder, Colab GPU, Wi-Fi
1555hrs–1715hrs	80 mins	LU4 — Machine Learning Implementation and	Demonstrations / Modelling, Drill and Practice	Trainer slides, Learner Guide, lab folder, Colab GPU, Wi-Fi

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		Operations · Topic 1: Build and benchmark generative solutions against state of the art. Hands-on (in class): Lab 21: Benchmark the Fine-Tuned Model Against State of the Art (slide 156). Topic 2: Recommend and implement bias mitigation and fairness interventions. Topic 3: Apply interpretability and explainability methods. Topic 4: Deploy and operationalize fine-tuned models — merging, GGUF export, deployment targets, continuous pre-training and the model card (trainer demonstration). Self-paced extension: Labs 22 –26.		
1715hrs–1730hrs	15 mins	Course recap: the ten key takeaways and the fine-tuning pipeline end to end; continuing your learning, Q&A and course feedback	Interactive presentation, Discussion	Trainer slides, TV, Whiteboard

Total instructional time: 450 minutes (7.5 hours), 9:30am – 5:30pm, plus one 30-minute lunch break.

Schedule Summary

Day	Theme	Instructional hours
Day 1	Fine tuning an open source LLM end to end — data engineering, data preparation, LoRA / QLoRA / Unsloth fine tuning, evaluation and deployment	7.5 hours
Total	**9:30 am – 5:30 pm, one 30-minute lunch break**	**7.5 hours**

Lab Reference

Lab	Title	Learning Unit	Delivery	Slide
1	Set Up the Fine-Tuning Environment on	LU1	In class	45

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	Colab			
2	Create a Versioned Data Repository for Fine Tuning	LU1	Self-paced	48
3	Generate Synthetic Training Data	LU1	Self-paced	51
4	Ingest Datasets from Multiple Sources	LU1	Self-paced	54
5	Transform Data into Chat and Instruction Formats	LU1	In class	57
6	Compare Generative Architectures and Accelerator Options	LU1	Self-paced	60
7	Clean and Sanitize a Fine-Tuning Dataset	LU2	In class	78
8	Feature Engineering for Fine Tuning	LU2	Self-paced	81
9	Build an ETL Pipeline for Fine-Tuning Datasets	LU2	Self-paced	84
10	Analyse and Visualise a Fine-Tuning Dataset	LU2	Self-paced	87
11	Tune Decoding Parameters for Higher-Quality Output	LU2	Self-paced	90
12	Transfer Learning, Distillation and Self-Supervised Learning	LU2	Self-paced	93
13	Frame the Problem and Select the Adaptation Method	LU3	Self-paced	115
14	Build LoRA from First Principles	LU3	Self-paced	118
15	Fine-Tune with PEFT LoRA	LU3	In class	121
16	Fine-Tune with QLoRA in 4-Bit	LU3	Self-paced	124
17	Accelerate Fine Tuning with Unsloth	LU3	In class	127
18	Monitor Training Stability and Convergence	LU3	Self-paced	130
19	Hyperparameter Optimization for LoRA	LU3	Self-paced	133
20	Advanced Loss Functions and	LU3	Self-paced	136

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	Generative Metrics			
21	Benchmark the Fine-Tuned Model Against State of the Art	LU4	In class	156
22	Detect and Mitigate Bias in Generative Outputs	LU4	Self-paced	159
23	Interpretability and Explainability for Generative Models	LU4	Self-paced	162
24	Intermediate Fine Tuning and Continuous Pre-Training	LU4	Self-paced	165
25	Merge, Export and Deploy the Fine-Tuned Model	LU4	Self-paced	168
26	Capstone — End-to-End Fine-Tuning Project	LU4	Self-paced	171

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